



JAIPUR NATIONAL UNIVERSITY

Semester/Year End/Back Examination: May 2024

Name of School:	School of Engineering and Technology
Name of Programme / Course:	B.Tech. (Biomedical) / B.Tech. (Biotech) / B.Tech. (Chemical) / B.Tech. (Civill) / B.Tech (Mechanical) / B.Tech. (CSE) / B.Tech. (CSE-AIML) / B.Tech. (CSE-Cyber Security) / B.Tech.(Foodtech) / B.Tech. (Electrical)
Semester/Year:	2nd
Paper Code:	BTEPPCO201T22
Title of Paper:	Programming for Problem Solving
Time: 3 hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

Attempt any five Questions from section A and attempt any four Questions from section B

Question
Number

SECTION-A

Marks
Allotted

1. Write a C program that reads from the user an arithmetic operator and two operands, perform the corresponding arithmetic operation on the operands using switch statement. 4
2. Show how break and continue statements are used in a C program, with example. 4
3. What are basic data types available in C? Write the significance of each data type. 4
4. Write a C program to read N integers into an array A and to find the (i) sum of odd numbers, (ii) sum of even numbers, (iii) average of all numbers. Output the results computed with appropriate headings. 4
5. Write a C program to find the area and perimeter of a circle. 4

- 6/ What are actual parameters and formal parameters? Illustrate with example. (3) 4
- 7/ What is the difference between structure and union? Explain the C syntax of structure declaration with example. (2) 4

SECTION-B

- 1/ What is an array? Explain the declaration and initialization of one dimensional and two dimensional array with an example. (2) 10
- 2/ Explain recursion, and write a program to find nth term of Fibonacci series. (3) 10
3. Give the scope and life time of the following:
(i) External variable (ii) Static variable (iii) Automatic variable (iv) Register variable 10
4. Explain with syntax and example, the different string manipulation library functions with example. 10
- 5/ Answer the following.
- i) What is pointer? Explain how the pointer variable declared and initialized? (2) 10
- ii) Write a C program to swap two numbers using call by address (pointers or reference) method. (4)